



BLACKBERRY HORIZON

SECURING INDUSTRIAL DIGITAL TWINS IN A SOFTWARE DEFINED FUTURE

Name: Jiyuan LiuUnikey: jliu0665DESN9002



KEY CHALLENGES

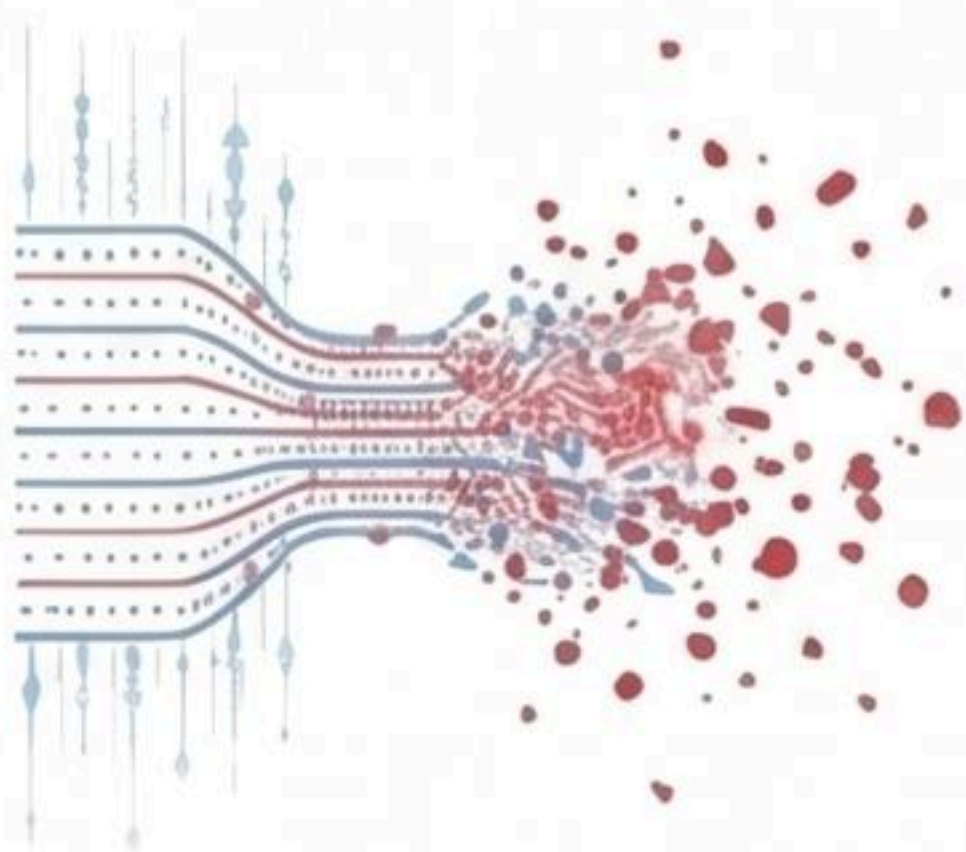


BACKGROUND





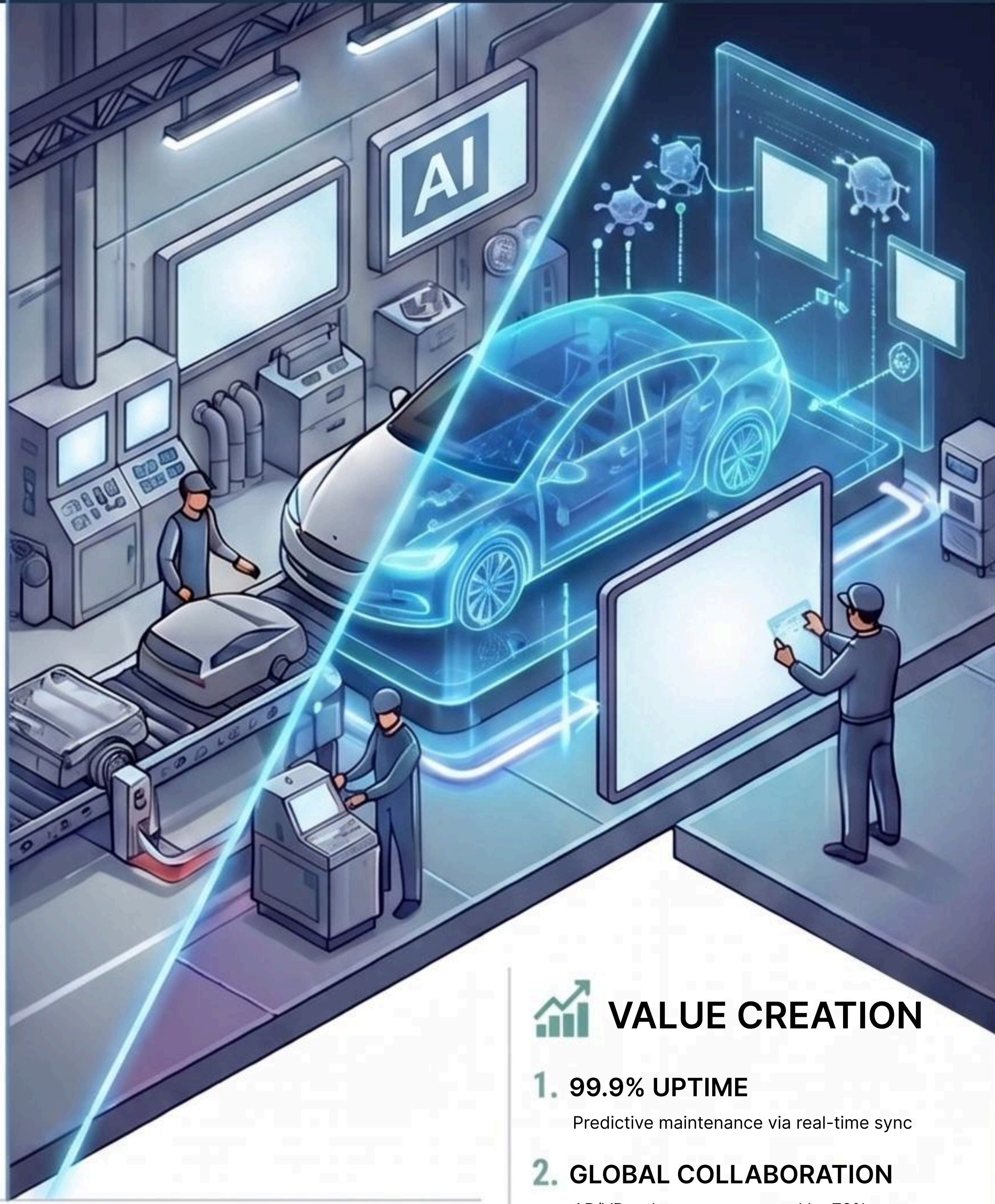
PAIN POINTS



1. OT/IT GAP:
Fragmented production vs digital models
2. REMOTE INEFFICIENCY:
Maintenance complexity requires on-site experts
3. SECURITY RISK:
Expanded IoT attack surface



THE SOLUTION





VALUE CREATION

1. 99.9% UPTIME
Predictive maintenance via real-time sync
2. GLOBAL COLLABORATION
AR/VR reduces expert travel by 70%
3. SECURE BY DESIGN
Defends edge data and intellectual property

IMPLEMENTATION PLAN

- Phase 1: Recruit 20+ AI/VR Experts
- Phase 2: Launch Pilot in key EV factory



PREDICT

BlackBerry QNX-driven sensors gather real-time vibrational and thermal telemetry from physical assets (e.g., robotic arms). Our generative AI models analyze this data stream, identifying pre-failure patterns and predicting critical component downtime 24 hours in advance with 99.9% accuracy.



NEUTRALISE

The moment a risk is predicted, our edge-computing node activates the BlackBerry Sentinel AI. This creates a secure "virtual patch" to isolate the specific code-base vulnerability, prevent the risk from propagating, and automatically reschedule production loads away from the affected machine to a virtual asset in the Metaverse.



HEAL

The physical asset undergoes human repair via an AR-guided remote maintenance interface, linked to the Digital Twin. The system then self-optimizes, not only confirming the physical fix but also automatically updating the underlying AI model and Digital Twin definition based on the newly repaired asset's performance, preventing repeat failures.